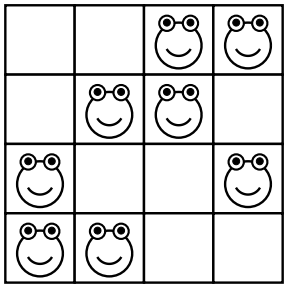
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 26588

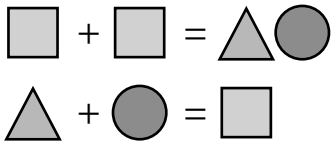
Part A: Each question is worth 3 points

1. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Only **one** frog jumps. How many different jumps are possible? (Count every frog and every empty field it can reach.)



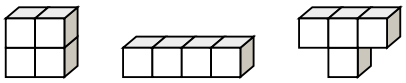
A. 8 ***B.*** 12 ***C.*** 14 ***D.*** 15 ***E.*** 16

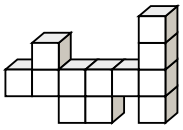
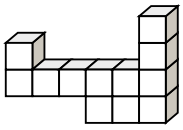
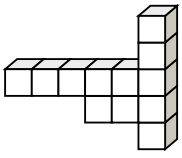
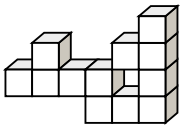
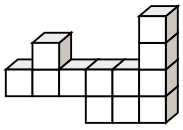
2. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Ben lays out the cards so that the calculations are correct. Which digit is hidden under the **circle**?



A. 1 ***B.*** 2 ***C.*** 7 ***D.*** 8 ***E.*** 9

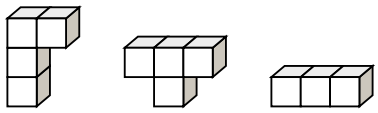
3. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures **can** he make?

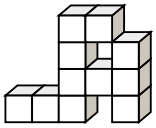
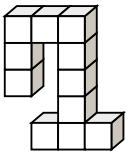
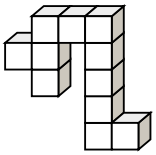
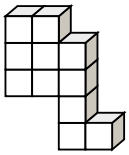
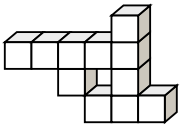


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part B: Each question is worth 4 points

4. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures can he **not** make?

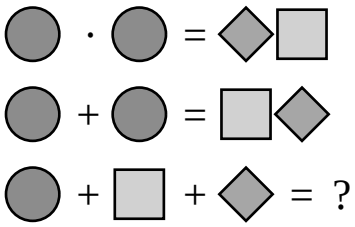


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

5. 100 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: black, dotted, striped, white, black, dotted, ... The cards repeat like this: star, triangle, diamond, star, triangle, diamond, ... What does child number **75** wear?

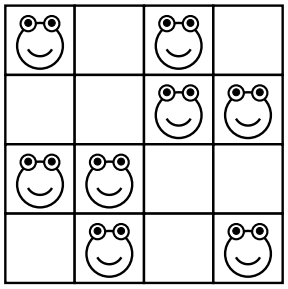
A. striped T-shirt, diamond card ***B.*** striped T-shirt, triangle card ***C.*** white T-shirt, diamond card
D. striped T-shirt, star card ***E.*** white T-shirt, star card

6. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Sara lays out the cards so that the calculations are correct. What do you get if you add the digits of all three shapes?



A. 17 ***B.*** 18 ***C.*** 19 ***D.*** 20 ***E.*** 22

7. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Two of the frogs jump at the same time, each to an empty neighbouring field. After that there are again exactly 2 frogs in each row and in each column. How many possibilities are there for two frogs to jump like this?



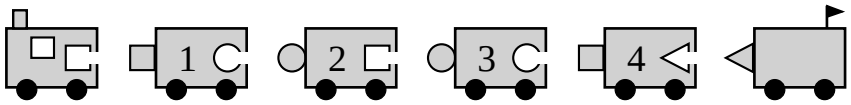
A. 1 ***B.*** 8 ***C.*** 12 ***D.*** 13 ***E.*** 24

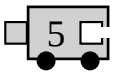
8. 43 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: white, striped, white, striped, ... The cards repeat like this: cross, triangle, diamond, square, star, cross, triangle, ... How many children have a **white** T-shirt **and** a **cross** card?

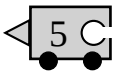
A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 8 ***E.*** 21

Part C: Each question is worth 5 points

9. Ben has these pieces: an engine, a last wagon and 4 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3, 4 or 5 numbered wagons, then the last wagon. Each piece can be used only once. Now Ben can build 3 trains. Ben gets one more wagon (number 5) and then can build exactly 8. Which wagon did Ben get?



A. 

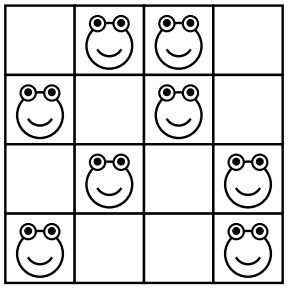
B. 

C.

D.

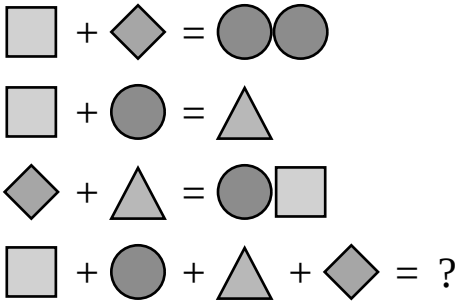
E.

10. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Two frogs jump at the same time, each to an empty neighbouring field, and after that there are again 2 frogs in each row and in each column. How many of the frogs can **never** be one of the two jumping frogs?



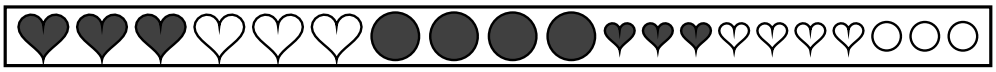
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 8

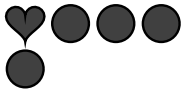
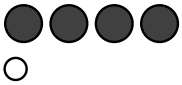
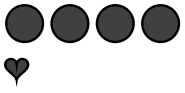
11. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Anna lays out the cards so that the calculations are correct. What number does Anna get for the last calculation?



A. 13 ***B.*** 14 ***C.*** 15 ***D.*** 17 ***E.*** 24

12. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white.
Three girls take cookies from the tray, one after the other:
Lena takes all the **heart-shaped** cookies that are still on the tray.
Zoe takes all the **white** cookies that are still on the tray.
Emma takes all the **large** cookies that are still on the tray.
One of the girls takes 4 cookies, one takes 6 and one takes 10. Which of these groups of cookies did one of the girls take?



A.  ***B.***  ***C.*** ***D.*** ***E.*** 

13. Anna has these pieces: an engine, a last wagon and 5 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3, 4 or 5 numbered wagons, then the last wagon. Each piece can be used only once. Anna has already put wagon 4 right after the engine. In how many different ways can Anna finish the train?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

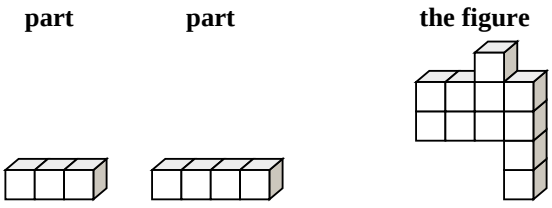
14. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Nora takes all the **heart-shaped** cookies that are still on the tray.
Ida takes all the **white** cookies that are still on the tray.
Emma takes all the **large** cookies that are still on the tray.
One of the girls takes 4 cookies, one takes 5 and one takes 6. How many cookies did **Nora** take?

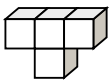
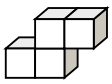
A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 7

15. 120 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: dotted, black, white, striped, dotted, black, ... The cards repeat like this: square, cross, diamond, star, circle, triangle, square, cross, ... How many children have a **dotted** T-shirt **and** a **cross** card?

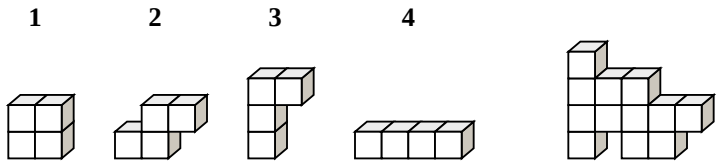
A. 0 ***B.*** 5 ***C.*** 10 ***D.*** 20 ***E.*** 30

16. Kanga built this flat figure out of 3 parts made of cubes, turned or flipped. Two of the parts are shown next to it. Which part is the third one?



A.  ***B.***  ***C.*** ***D.*** ***E.***

17. Kanga has these 4 parts made of cubes (numbered 1 to 4). He builds the flat figure on the right with **three** of them, turned or flipped. Which part is left over?

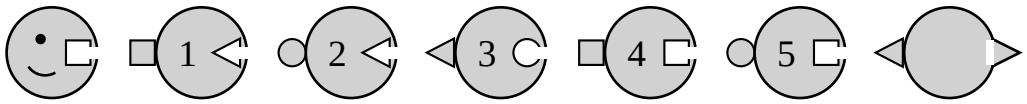


A. part 1 ***B.*** part 2 ***C.*** part 3
D. part 4 ***E.*** none, he needs all 4

18. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Tina takes all the **heart-shaped** cookies that are still on the tray.
Rosa takes all the **white** cookies that are still on the tray.
Nora takes all the **large** cookies that are still on the tray.
One of the girls takes 3 cookies, one takes 4 and one takes 9. How many cookies did **Tina** take?

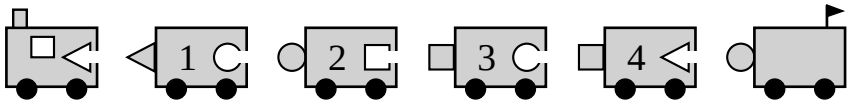
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

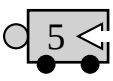
19. Tom has these pieces: a head, a tail and 5 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3, 4 or 5 numbered parts, then the tail. Each piece can be used only once. Tom has already put part 1 right after the head. In how many different ways can Tom finish the caterpillar?

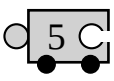


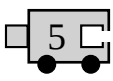
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

20. Leo has these pieces: an engine, a last wagon and 4 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3, 4 or 5 numbered wagons, then the last wagon. Each piece can be used only once. Now Leo can build 2 trains. Leo gets one more wagon (number 5) and then can build exactly 7. Which wagon did Leo get?



A. 

B. 

C. 

D.

E.

Answer Key

1.	**C**	**2.**	**D**	**3.**	**E**	**4.**	**C**	**5.**	**A**	**6.**	**B**
7.	**C**	**8.**	**C**	**9.**	**A**	**10.**	**A**	**11.**	**C**	**12.**	**C**
13.	**B**	**14.**	**C**	**15.**	**A**	**16.**	**E**	**17.**	**C**	**18.**	**D**
19.	**B**	**20.**	**E**								

Solutions

1. C (14) — Go frog by frog and count its empty neighbours: 1 + 1 + 3 + 2 + 2 + 3 + 0 + 2 = 14.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog]

2. D (8) — Two digits add up to at most 9 + 9 = 18, so the triangle at the front of the two-digit answer must be 1. Then try the possible digits and check both calculations: circle = 8, square = 9, triangle = 1.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · find_digit]

3. E — Every figure has 12 cubes, so count is not enough. Start with a cube that sticks out: only one part can fill it. Only one figure can be filled completely. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · can_make]

4. C — Try to fill each figure with the parts, starting at a corner or a cube that sticks out. Four figures can be filled; in figure C there is always a gap that no part fits. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · cannot_make]

5. A (striped T-shirt, diamond card) — The T-shirts repeat every 4: 75 = 18 × 4 + 3, so the T-shirt is the 3rd one: striped. The cards repeat every 3: 75 = 24 × 3 + 3, so the card is a diamond. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

6. B (18) — Try the possible digits (start with the multiplication: which times-table result has these digits?) and check every calculation: circle = 9, square = 1, diamond = 8. 9 + 1 + 8 = 18. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · times]

7. C (12) — If a frog jumps sideways, one column loses a frog and the next one gets one more - the other frog must jump sideways between the same two columns, the other way, in another row (the same for jumps up and down between two rows). Checking each pair of neighbouring columns and rows gives 12.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · count_pairs]

8. C (5) — The patterns start again together every 10 children; in each block child number 1 matches. 43 = 4 × 10 + 3: 4 full blocks give 4, and in the 3 children left over child 41 matches too. 4 + 1 = 5.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

9. A — Try each new piece and count again with a tree. With A: H–1–2–4–T, H–1–2–5–4–T, H–1–3–2–4–T, H–1–3–2–5–4–T, H–4–T, H–5–1–2–4–T, H–5–1–3–2–4–T, H–5–4–T, that is 8.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · extra_piece]

10. A (0) — Find all the ways (15): the second jump must undo the first one (sideways between the same two columns, or up/down between the same two rows). 8 frogs take part in some way, so 8 - 8 = 0 never jump.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · never_jumps]

11. C (15) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Start with the calculation where the fewest digits are possible, then check all three: square = 2, circle = 1, triangle = 3, diamond = 9. 2 + 1 + 3 + 9 = 15. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · four_cards]

12. C — Try the orders. Only Zoe, Lena and then Emma gives 4, 6 and 10: Zoe takes 10, Lena takes 6, Emma takes 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · which_group]

13. B (2) — Look only at the branch that starts with wagon 4: H–4–T, H–4–3–T. That is 2.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]

14. C (4) — Try the orders one by one. Only Emma, Ida and then Nora gives 4, 5 and 6. So Nora took 4.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · how_many_one]

15. A (0) — Write the first children down: both patterns start again together already after 12 children (not 4 × 6 = 24). In each block of 12 no child has a dotted T-shirt and a cross card, so the answer is 0.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]

16. E — The figure has 11 cubes and the two parts have 7, so the third part has 4 cubes. Put the two parts into the figure; what is left has the shape of part E. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · missing_part]

17. C (part 3) — The figure has 12 cubes and the parts have 16 together, so the left-over part has 4 cubes. Then check which three parts really fill the figure: part 3 is left over. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · left_over]

18. D (4) — Try the orders one by one. Only Nora, Rosa and then Tina gives 3, 4 and 9. So Tina took 4.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · how_many_one]

19. B (2) — Look only at the branch that starts with part 1: H–1–T, H–1–3–2–T. That is 2.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]

20. E — Try each new piece and count again with a tree. With E: H–1–T, H–1–2–3–T, H–1–2–4–5–3–T, H–5–3–T, H–5–3–2–4–1–T, H–5–4–1–T, H–5–4–1–2–3–T, that is 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · extra_piece]